

4 Managing extreme weather

Enquiry question: *How can we best respond to and cope with the impacts of extreme weather?*

What students need to learn	Suggested teaching and learning
■ Fieldwork and research into ways of managing and responding to extreme weather events using short- and longer-term strategies, and how some management strategies are more successful than others.	■ Using primary and secondary sources to investigate strategies such as USA hurricane warning, Environment Agency flood protection and risk assessments at a local scale, eg York.
■ The role of new technology in improving community preparedness, event forecasting and reducing impacts of disasters.	■ Researching the role of technology and its application to extreme weather management such as forecasting (NOAA), flood monitoring or the use of drought resistant crops.
■ Ways to manage drought through physical, social, economic and political responses in contrasting areas.	■ Assessing sustainable longer-term solutions for tackling drought such as water management and adapting farming techniques as in south east England or Ethiopia.

2.4 Topic 2: Crowded Coasts

Crowded coasts

The coastal zone is one of the most densely populated areas globally because of favourable locational factors, yet it is an area of immense environmental value. This topic allows students to carry out detailed studies of contrasting coastal environments at a local/small scale, exploring competing and conflicting land uses, and evaluating the pressures created by development. It considers how vulnerable and valuable coastlines face a growing physical risk from rapid coastal erosion and coastal flooding. It provides an overview of coastal management, looking at protection and conservation strategies for the present and the future.

Fieldwork and research opportunities

Fieldwork opportunities include surveys of coastal development and land use, coastal management, and the state of the coastal environment. Opportunities for **research work** include satellite images to show coastal change, maps to calculate coastal erosion, shoreline management and statistics for coastal retreat and flooding.

1 Competition for coasts

Enquiry question: *Why is the coastal zone so favoured for development?*

What students need to learn	Suggested teaching and learning
■ How physical factors create variety in a range of different coastal environments.	■ Examining the range of natural factors that influence coastal environments and their value, eg geology, geomorphology and ecosystems.
■ The factors which have led to exponential population growth in some coastal environments: ◆ flat land, soil fertility, equable climate, and biodiversity ◆ potential for fishing, recreation/tourism, industrial and port development and accessibility.	■ Developing awareness of the exponential population growth in accessible and popular coastal areas.
■ Fieldwork and research to show how these factors have shaped the development and growth of contrasting crowded coasts over time.	■ Investigating the growth of contrasting crowded coasts such as a UK resort coastline and a foreign one eg a Spanish Costa or the Florida coast.

2 Coping with the pressure

Enquiry question: *How do various coastal developments create competition and conflict?
How can these pressures be resolved?*

What students need to learn	Suggested teaching and learning
■ How development lead to patterns of zoning in coastal areas and how competition for space puts pressure on coastal environments.	■ Exploring: ◆ land use in a resort/port ◆ the need for planning control as the available land decreases and rate of development intensifies.
■ Fieldwork and research into the pressures on the coast when development and conservation meet head on including: ◆ the overuse of resources, pollution, other developments ◆ the destruction of high-value coastal habitats.	■ Using primary and secondary sources to investigate the pressures associated with coastal development, eg: ◆ analysing the impacts of fishing, aquaculture, marine and beach pollution and tourism ◆ assessing the value of and level of destruction in, eg sand dunes, salt marshes or SSSIs.
■ There are economic benefits and environmental costs to coastal development which influence the success of the development equation and involve the views of stakeholders and their conflicting needs.	■ Exploring a suitable case study eg Dibden Bay or Akamas (Cyprus) to: ◆ assess the beneficial economic impacts of development against the environmental costs (using, eg CBA and Environmental Impact As) ◆ examining the views and objectives of interested parties (using Conflict matrices and values analysis).